KYG - Know Your Genes

Phase 1 Website Specification

Target Launch: May 14, 2026

Competitor Reference: mapmygenome.in (Shopify-based)

1. Why Next.js Over Shopify

What Shopify Gives You

- Quick storefront setup
- Built-in payment gateway
- Basic product catalog and checkout
- Hosting included

What Shopify Cannot Do (Why KYG Needs Next.js)

Capability	Shopify	Next.js
User Dashboard	Not possible. No concept of a logged-in user dashboard with history, reports, preferences.	Full control. Build any authenticated experience - report history, kit tracking, health timelines, loyalty points.
Role-Based Access	Only store admin roles. No custom roles (user, counsellor, admin, lab-partner).	Implement any RBAC model with middleware-level route protection.
Shiprocket Integration (Forward + Reverse)	Requires third-party Shopify app (limited, paid, no reverse-pickup control).	Direct Shiprocket API integration. Full control over forward delivery of kit AND reverse pickup of sample.
Report Delivery System	No way to deliver genetic test reports inside the platform. You'd email PDFs separately.	Reports rendered in-app, tied to user account, with history and doctor/counsellor annotations.
Genetic Counsellor Booking	Not native. Need Calendly embed or external tool.	Built-in booking system tied to user profile and test results.
Quiz / Risk Assessment	Basic Shopify quiz apps exist but are shallow and don't integrate with your data.	Custom quiz logic that feeds into product recommendations and stores results in user profile.
Loyalty & Rewards	Third-party apps with monthly fees, limited customization.	Custom loyalty card system integrated with purchase history and referrals.
Blog with SEO Control	Basic blog, limited SEO flexibility.	Full MDX/CMS-driven blog with schema markup, OG tags, programmatic SEO.
Multi-stage User Journey	Shopify is transactional (buy and leave).	Lifecycle engagement: education -> purchase -> kit tracking -> sample -> report -> counselling -> follow-up.
Partner/B2B Portal	Not feasible.	Dedicated partner-with-us flow, B2B dashboards for clinics/hospitals.
Government Program Integration	Not possible.	Custom flows for subsidized testing, bulk orders, program-specific reporting.
Data Privacy & Compliance	Data lives on Shopify's servers (US/Canada). Genetic data has strict Indian regulations.	Self-hosted or India-region cloud. Full control over where genetic data is stored.
Cost at Scale	Shopify Plus starts at $2,000+/month. Transaction fees on every order.	Vercel/AWS hosting: fraction of the cost. No per-transaction platform fees.

The Core Argument

MapMyGenome uses Shopify because they treat genetic testing as a product you buy. KYG's value proposition is different - it's a health relationship platform where the test kit purchase is just the

entry point. The real product is the ongoing dashboard, reports, counselling, literacy content, and lifecycle engagement. Shopify cannot be this platform. Next.js can.

2. Recommended Tech Stack (Phase 1)

Layer	Technology	Reason
Framework	Next.js 15 (App Router)	SSR + SSG, API routes, middleware auth, ISR for blog/content
Language	TypeScript	Type safety across the full stack
Styling	Tailwind CSS + shadcn/ui	Rapid UI development, consistent design system
Auth	NextAuth.js (Auth.js v5)	Google/Email login, role-based sessions, JWT
Database	PostgreSQL (Supabase or Neon)	Relational data for users, orders, reports, roles
ORM	Prisma	Type-safe database queries
Payments	Razorpay	Indian payment gateway, UPI/cards/netbanking
Shipping	Shiprocket API	Forward + reverse logistics for kit delivery
CMS (Blog)	MDX files or Sanity CMS	Content team can publish without deploys
Email	Resend or AWS SES	Transactional emails (order confirmation, report ready)
Hosting	Vercel	Zero-config Next.js deployment, edge functions, India CDN
Storage	AWS S3 (Mumbai region)	Report PDFs, user documents - data stays in India

3. Phase 1 - Complete Page Map

3.1 Marketing Pages (Public)

#	Route	Purpose	Key Elements
1	/	Homepage	Hero, value props, featured packages, trust signals, CTA
2	/how-it-works	4-step process	Order -> Receive Kit -> Collect Sample -> Get Report
3	/why-kyg	Differentiators	Indian-specific reference data, affordability, privacy-first, counsellor access
4	/science	Scientific credibility	Lab certifications, methodology, accuracy stats, research backing
5	/about	Company story	Mission, team, advisors, lab partners
6	/contact	Contact form + info	Form, email, phone, office address, Google Maps embed
7	/press	Media coverage	Press releases, media mentions, downloadable press kit
8	/quiz	Risk/recommendation quiz	Interactive quiz -> personalized package recommendation
9	/partner-with-us	B2B lead capture	For clinics, hospitals, corporate wellness, government programs

3.2 Commerce Pages (Public + Auth)

#	Route	Purpose	Key Elements
1	/packages	Product catalog	All test packages with pricing, comparison table
2	/packages/[slug]	Package detail	Description, biomarkers, sample report, FAQ, add-to-cart
3	/cart	Shopping cart	Line items, quantity, promo code, price summary
4	/checkout	Checkout flow	Address (Shiprocket pin check), payment (Razorpay), order summary
5	/checkout/success	Order confirmation	Order ID, expected delivery, next steps
6	/track/[orderId]	Order tracking	Shiprocket tracking - kit delivery + reverse pickup status

3.3 Content Pages (Public)

#	Route	Purpose	Key Elements
1	/blog	Blog listing	Categories: Genetic Literacy, Wellness, Research, Stories
2	/blog/[slug]	Blog post	Full article, author, share buttons, related posts, CTA
3	/faq	FAQ	Accordion-style, categorized (Testing, Privacy, Reports, Shipping)
4	/counsellor	Genetic counselling	What is counselling, counsellor profiles, booking CTA

3.4 Legal Pages (Public)

#	Route	Purpose
1	/privacy	Privacy policy - critical for genetic data trust
2	/terms	Terms of service
3	/cookies	Cookie policy

3.5 Auth Pages

#	Route	Purpose	Key Elements
1	/login	User login	Email/password, Google OAuth, OTP option
2	/signup	User registration	Name, email, phone, password, terms acceptance
3	/forgot-password	Password reset	Email input -> reset link -> new password

3.6 User Dashboard (Authenticated)

#	Route	Purpose	Key Elements
1	/dashboard	Overview	Active orders, recent reports, upcoming counselling, loyalty points
2	/dashboard/orders	Order history	All orders with status, tracking links, reorder option
3	/dashboard/orders/[id]	Order detail	Full order info, Shiprocket tracking, kit instructions
4	/dashboard/reports	Report history	All genetic test reports, download PDF, view in-app
5	/dashboard/reports/[id]	Individual report	Full report with explanations, actionable insights
6	/dashboard/profile	Profile management	Personal info, health profile, communication preferences
7	/dashboard/family	Family history	Family tree builder, hereditary risk flags
8	/dashboard/loyalty	Loyalty / rewards	Points balance, earn/redeem history, referral link
9	/dashboard/bookings	Counsellor appointments	Book, reschedule, cancel counselling sessions

3.7 Admin Panel (Role: Admin)

#	Route	Purpose
1	/admin	Admin dashboard - order stats, revenue, user counts
2	/admin/orders	Manage all orders, update status, trigger Shiprocket actions
3	/admin/users	User management, role assignment
4	/admin/reports	Upload/manage genetic reports for users
5	/admin/content	Blog post management
6	/admin/packages	Manage test packages, pricing, availability
7	/admin/coupons	Promo codes and discount management

4. Shiprocket Integration Flow

Forward Flow (Kit to Customer)

- Order Placed
- Create Shiprocket Order (POST /orders/create)
- Generate AWB
- Kit Shipped
- Customer Receives Kit
- Status: 'Kit Delivered'

Reverse Flow (Sample back to Lab)

- Customer Collects Sample
- Request Reverse Pickup via Shiprocket API (POST /orders/create/return)
- Pickup Scheduled
- Sample Collected
- Sample Reaches Lab
- Status: 'Sample Received'
- Lab Processing (3-4 weeks)
- Report Ready -> Notify User

Key Shiprocket APIs

- POST /orders/create - forward shipment
- POST /orders/create/return - reverse pickup
- GET /courier/track/shipment/{id} - tracking
- GET /courier/serviceability - pin code check at checkout

5. User Roles

Role	Access
Guest	Marketing pages, blog, FAQ, quiz, package browsing
User	Everything above + dashboard, orders, reports, bookings, loyalty
Counsellor	User access + counsellor dashboard, patient report access, session management
Admin	Full access including admin panel, order management, report upload, user management

6. Phase 1 Priority (MVP for May 14, 2026 Launch)

Must Have (Launch Blockers)

- Homepage with clear value proposition
- How It Works page
- Package catalog + detail pages (minimum 3 packages)
- Cart + Checkout with Razorpay
- Shiprocket forward delivery integration
- User signup/login (email + Google)
- User dashboard with order tracking
- Basic blog (5-10 genetic literacy articles for SEO)
- Contact page
- Privacy policy + Terms
- Mobile responsive design
- Admin: order management + report upload

Should Have (Week 1-2 Post-Launch)

- Quiz / recommendation engine
- Shiprocket reverse pickup integration
- Report viewing in dashboard
- FAQ page
- Coupon/promo code system

Nice to Have (Phase 2)

- Loyalty card system
- Counsellor booking
- Family history builder
- Partner portal
- Service add-ons (Cult Fit, Paytm tie-ups)
- Age-by-stages journey mapping
- Government program flows
- NRI-specific packages and shipping

7. Target User Segments

Segment	Key Need	Entry Point
Pre-Conception Couples	Carrier screening, hereditary risk	Quiz, Why KYG, packages
Family History Concerns	Cancer/cardiac/diabetes risk assessment	Science page, counsellor
Chronic Care Patients	Pharmacogenomics, treatment optimization	Doctor referral, partner portal
NRI Segment	Indian-specific reference data, family testing	Direct marketing, premium packages
Bio-Hackers / Young Parents	Wellness optimization, proactive health	Blog, quiz, social media

8. Key Differentiators to Highlight on Website

- Indian-Specific Reference Data - Not Western averages applied to Indian genetics
- Affordable - Lower barrier to entry, EMI options, government program alignment
- Privacy-First - Data stored in India, user controls data, no third-party sharing
- Actionable Reports - Not just data dumps, clear next steps with counsellor support
- Genetic Literacy - Education-first approach builds trust before selling
- No Marriage Stigma - Sensitive handling of carrier status, counselling-first approach
- Clinical Grade - Not just ancestry/wellness fluff, actual clinical utility